

Note on Methodological Minimalism in Quantum Foundations

Abstract

This note argues that several long-standing interpretational problems in quantum mechanics persist not because they require additional postulates, entities, or metaphysical commitments, but because their structural core has not been isolated with sufficient clarity. We advocate **methodological minimalism**: the principle that foundational explanations should introduce no assumptions beyond those already present in the mathematical structure of the theory. We show that key phenomena—most notably measurement and collapse—can be understood as consequences of spectral structure and boundary constraints, without invoking ontological randomness, consciousness, or non-physical mechanisms. Simplicity here is not aesthetic; it is a diagnostic indicator of structural correctness.

1. Minimalism as a Method, Not an Ideology

Methodological minimalism is often misunderstood as a preference for fewer equations or reduced formalism. This is incorrect. Minimalism concerns **assumptions**, not symbols.

A theory is methodologically minimal if:

1. Every postulate is operationally or structurally necessary.
2. No explanatory entity is introduced without independent functional role.
3. All interpretations remain internal to the formalism already in use.

In this sense, minimalism is an extension of Occam's razor applied to *ontology*, not to mathematics.

2. The Pathology of Excess Assumptions in Quantum Foundations

Quantum mechanics is mathematically economical but interpretationally overloaded. Standard accounts of measurement typically add at least one of the following:

- Ontological randomness,

- Observer-dependent collapse,
- Many-world branching,
- Hidden variables external to the Hilbert space.

These additions share a common feature: they are **not required by the linear-algebraic structure of the theory itself**. They compensate for an unresolved reading of that structure rather than extending it.

3. Spectral Structure Is Already Doing the Work

The core mathematical objects of quantum mechanics—Hermitian operators, inner products, spectral decompositions—already determine:

- the admissible states,
- the stable modes of evolution,
- the projection structure associated with measurement.

Eigenvectors are not arbitrary mathematical conveniences; they are the only directions in state space that remain invariant under the action of the operator. Any account of measurement that treats eigenstates as ontologically emergent from nothing introduces unnecessary metaphysics.

Methodological minimalism insists on the opposite reading: **if the structure selects them, they pre-exist as structural possibilities**.

4. Boundary Conditions as Explanatory Primitives

A minimal framework recognizes boundary conditions—not observers—as the relevant source of state reduction. Boundaries are already part of physical modeling: experimental setup, environmental coupling, device constraints.

Once boundaries are treated as operators inducing effective metrics or generators, collapse becomes a case of **spectral selection under constraint**, not a stochastic miracle.

No new physics is added. No axioms are violated. Nothing external to the formalism is invoked.

5. Why Simplicity Signals Depth

Historically, foundational advances simplify rather than complicate:

- Newton reduced celestial and terrestrial motion to the same equations.
- Maxwell unified electricity and magnetism through field structure.
- Quantum mechanics itself replaced classical complexity with linear operators.

In each case, resistance came not from mathematical inadequacy, but from conceptual inertia. Methodological minimalism often appears trivial *after* it is accepted, and intolerably reductive *before*.

6. A Diagnostic Criterion

We propose the following diagnostic criterion for foundational proposals:

If a phenomenon can be explained entirely within the spectral and boundary structure already present in the theory, then any additional ontological postulate is methodologically unjustified.

This criterion does not prohibit extensions. It simply demands necessity.

7. Conclusion

Quantum mechanics does not suffer from a lack of interpretation; it suffers from an excess of them. Methodological minimalism offers a corrective by returning attention to the internal structure of the theory and asking a disciplined question: *What is already doing the work?*

When the answer is “the spectrum,” “the boundary,” or “the operator itself,” the appropriate response is not to add metaphysics, but to remove it.

Clarity, in this context, is not a stylistic virtue. It is evidence that the structure has finally been seen.

This note is intended as a methodological clarification, not as a competing interpretation. Its claim is narrower and stronger: many interpretational problems dissolve when the formal structure of quantum mechanics is taken seriously enough—and nothing more is added.

Note on Ontological Priority and Scope

Purpose of this Note

This note serves a purely **registral and clarificatory function**. It does not advance new results, rebut alternative interpretations, or claim exclusivity over mathematical techniques that are part of the standard toolkit of physics. Its sole purpose is to **state explicitly the ontological reading and scope** under which the present framework has been developed.

Ontological Claim

The central ontological claim articulated in this work is the following:

Measurement outcomes and stable states do not arise through ontological creation or irreducible randomness; they correspond to the projection of pre-existing structural possibilities onto an observable domain under boundary constraints.

- This claim is formulated explicitly in terms of a distinction between:
 - a latent structural domain (denoted in earlier versions as $\mathcal{N}_1$),
 - an observable domain ($\mathcal{N}_0$),
 - boundary-induced operators mediating the projection between them.

Eigenstates are therefore interpreted not as post-measurement artifacts, but as **structurally admissible modes that pre-exist as possibilities prior to observation**.

Distinction from Prior Literature

While spectral decompositions, decoherence theory, and pointer-state analyses are well established in the literature, their use has traditionally been **instrumental rather than ontological**. In contrast, the present framework assigns **ontological status** to the spectral structure itself:

- Spectral modes are treated as *ontological deltas* (structural anchors),
- Eigenvalues as necessity weights,
- Collapse as boundary-induced spectral selection rather than stochastic postulate.

This shift concerns **interpretation and unification**, not mathematical novelty.

Priority Statement

The explicit formulation of quantum measurement and state selection as a **boundary-induced projection from a latent structural space of possibilities into realized states**, without recourse to observer-dependence, consciousness, or ontological randomness, was articulated in the publicly timestamped versions of this work released in January 2026.

This statement of priority concerns the **ontological framing and unifying principle**, not the underlying linear-algebraic tools themselves.

Scope and Intent

The framework is intended to be:

- compatible with standard quantum mechanics,
- compatible with Born's rule and Lindblad dynamics,
- non-competitive with decoherence theory,
- minimal in ontological commitments.

Any subsequent reformulation, reinterpretation, or application of these ideas within alternative vocabularies or formalisms should be understood as operating within the conceptual space delineated here.

Closing Remark

This note is offered in the spirit of scientific clarity and continuity. Foundational progress benefits not only from new results, but from precise identification of **what is being claimed, at what level, and with what ontological commitments**.

No stronger claim is made.